Does domain-general or domain-specific cognitive capacity modulate syntactic prediction?

Readers engage in predictive processing to facilitate English sentence processing[1]. Staub & Clifton[2] found readers utilise "either" to generate syntactic prediction, leading to shorter reading times at the "or" coordinating structure. Previous studies had mixed results on the modulating factors of predictive processing. Huettig & Janse[3] suggested working memory (WM) and processing speed (PS) predict semantic prediction, while Hintz et al.[4] suggested PS as the sole predictor. Recent studies proposed dissociable verbal WMs, including phonological WM (PWM) and semantic WM (SeWM)[5], but whether the verbal WMs modulate predictive processing distinctly remains unclear. This study provides a holistic investigation of the modulating factors of predictive processing by adopting an individual differences approach.

We tested whether readers engage in syntactic prediction while comprehending "either…or" coordinating sentences[2], followed by measurement of **PWM, SeWM**, spatial WM (**SpWM**), and **PS** (*M* accuracies = 90%, 92%, 76%, 75%, respectively) (**Table 1**). Native English speakers (*N* = 86) read 22 target sentences in which predictability was manipulated by varying the presence of "either." In the predictable condition, they read "Either Congress passed the bill or the President issued an order." In the unpredictable condition, they read sentences without "either", while other parts remained identical. Readers answered comprehension questions after 31% of the trials (*M* accuracy = 98%). We analysed the **"Or NP"** (e.g., or the President) as the **critical** region, and the words following it until the sentence-final word (e.g., issued an order.) as the **spillover** region, following Staub & Clifton[2]. Linear-mixed effect models investigated the main effects of predictability, PWM, SeWM, SpWM, PS, as well as 2-way interactions of predictability by PWM, predictability by SeWM, predictability by SpWM, and predictability by PS on first pass and go past reading times. The main effect of trial order, and the 2-way interaction of predictability by trial order on first pass and go past reading times were included as covariates.

The eye-tracking results replicated the previous result[2] (**Fig. 1**). In the **"Or NP"** region**,** first pass (281 ms vs. 300 ms), β = -.08, *SE* = .02, *t* = -3.07, and go past times (335 ms vs. 377ms), β = -.07, *SE* = .03, *t* = -2.20 were shorter in the predictable (vs. unpredictable) conditions. In the **spillover** region, first pass time was longer in the predictable (vs. unpredictable) conditions (579 ms vs. 540 ms), but did not reach statistical significance, β = -.04, *SE* = .03, *t* = 1.35. Go past time was shorter in the predictable (vs. unpredictable) conditions (1183 ms vs. 1248 ms), β = -.07, *SE* = .03, *t* = -2.05. In the **individual differences analysis on the "Or NP" region**, results revealed a significant main effect of PS on first pass, β = -.07, *SE* = .02, *t* = -2.97, and go past times, β = -.07, *SE* = .03, *t* = -2.16. Additionally, there was a near significant interaction of predictability x PS on first pass time, β = -.02, *SE* = .01, *t* = -1.83 (**Fig. 2**). In the **individual differences analysis on the spillover region**, results revealed a significant main effect of PS on first pass, β = -.10, *SE* = .03, *t* = -2.78, and go past times, β = -.19, *SE* = .06, *t* = -2.85. No other main effects and interactions were significant. Post-hoc analysis revealed low reliability of predictability effects from split-half correlations (e.g., first pass time; *r* = -0.14, and go past time on critical region; *r* = 0.09).

The predictability effect on **first pass** time on the critical region suggests easier initial processing with "either" present. The predictability effect on **go past** time on the critical and spillover regions suggests less frequent reanalysis with "either" present. The main effect of **PS** on reading measures suggests that readers with higher PS generally read faster, which may be supported by faster sub-processes of sentence processing (e.g., word recognition, syntactic analysis, and accessing real-world knowledge). The near significant interaction of predictability x **PS** on **first pass** time in the critical region suggests readers with higher PS generally read faster, facilitating more accurate retention of the syntactic property of "either" (i.e., coordinating two sentences). With a more accurate representation of "either," readers with higher PS may more strongly predict S-coordination. These findings suggest that syntactic prediction may be supported by domain-general cognitive capacity.

Table 1. Individual difference measurements. The dependent variables were z-transformed for individual difference analysis

Construct	Task	Description	Dependent variable
Phonological WM	Word span task	Participants recall English words or nonwords with varying number of words or syllables	% words recalled correctly
	Nonword span task		% phonemes recalled correctly
Semantic WM	Category probe task	Judging whether words are of the same category	% of correct trials
	Synonym judgement task	Judging whether words are synonyms	
Spatial WM	Shortened symmetry span task	Judging the symmetry of pictures while memorising square positions	The sum of the recall correct
	Shortened rotation span task	Judging the rotation of letters while memorising arrows	
Processing speed	The number-symbol task	Copying symbols corresponding to numbers	% of correct trials
	Letter comparison task	Judging whether letter strings are identical	

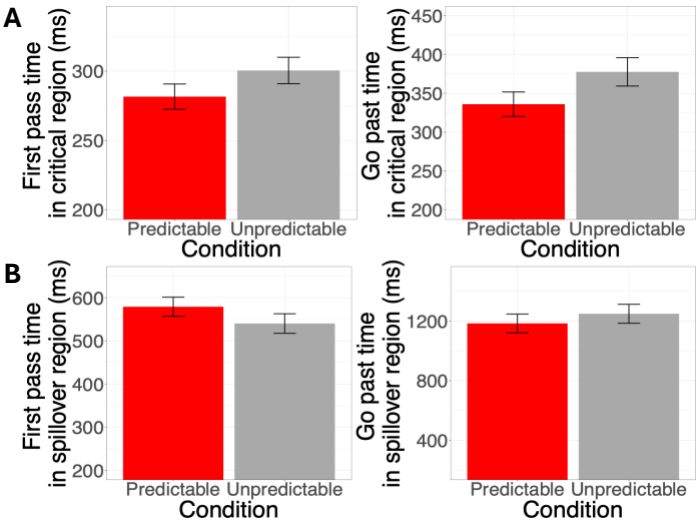

Fig. 1 (A) Mean first pass and go past times in the critical region, and **(B)** Mean first pass and go past times in the spillover region. Error bars indicate 95% confidence intervals.

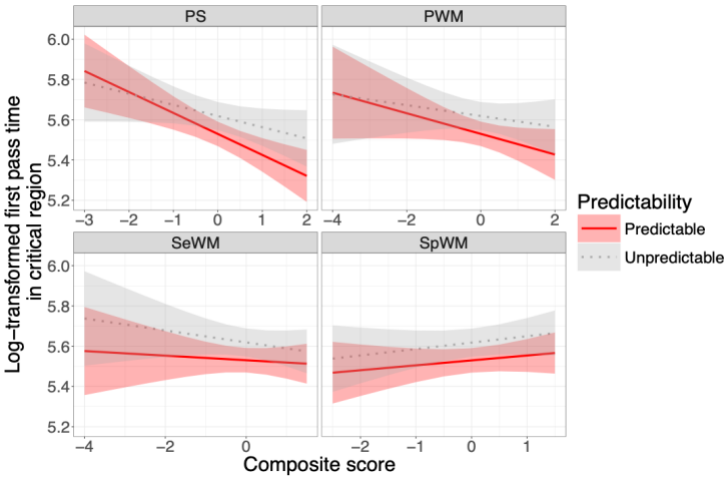

Fig. 2. The estimated marginal means from the model testing the 2-way interaction of predictability by individual difference predictors on **first pass time**.

References

[1] Kuperberg & Jaeger (2015). *Lang. Cogn. and Neurosci*. [2] Staub & Clifton (2006). *J. Exp. Psychol. Learn. Mem. Cogn.* [3] Huettig & Janse (2016). *Lang. Cogn. and Neurosci*. [4] Hintz et al. (2024). *Sci Rep.* [5] Martin et al. (2025). *Cogn. Affect. Behav. Neurosci.*